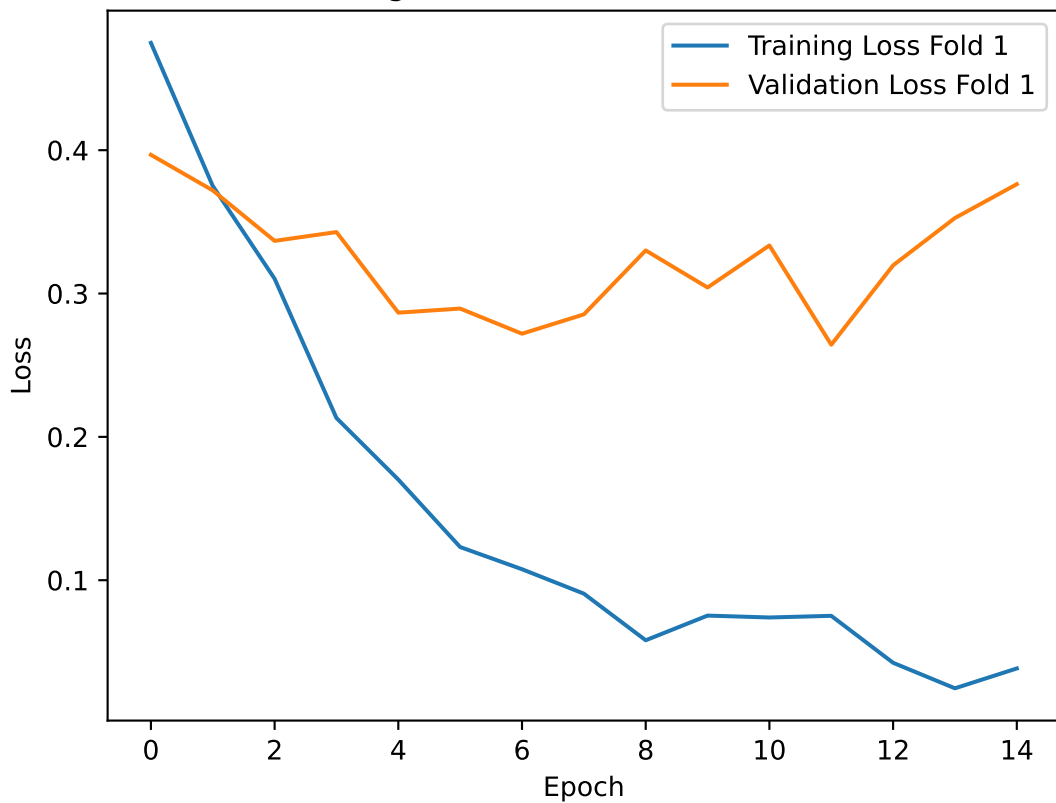


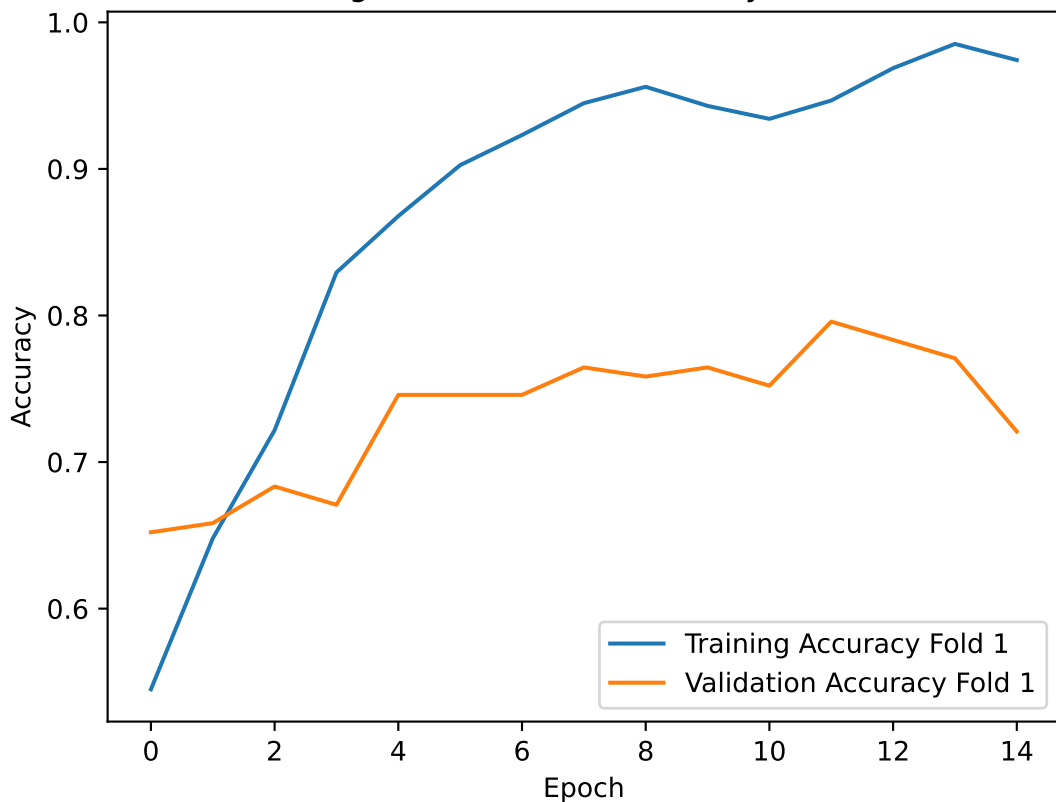
Validation F1-Score for fold 1: 0.03787878787878787

0.03787878787878787

Training and Validation Loss for Fold 1



Training and Validation Accuracy for Fold 1



Best Hyperparameters for Fold 1:

```
{'conv_filters': 64, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 160}
```

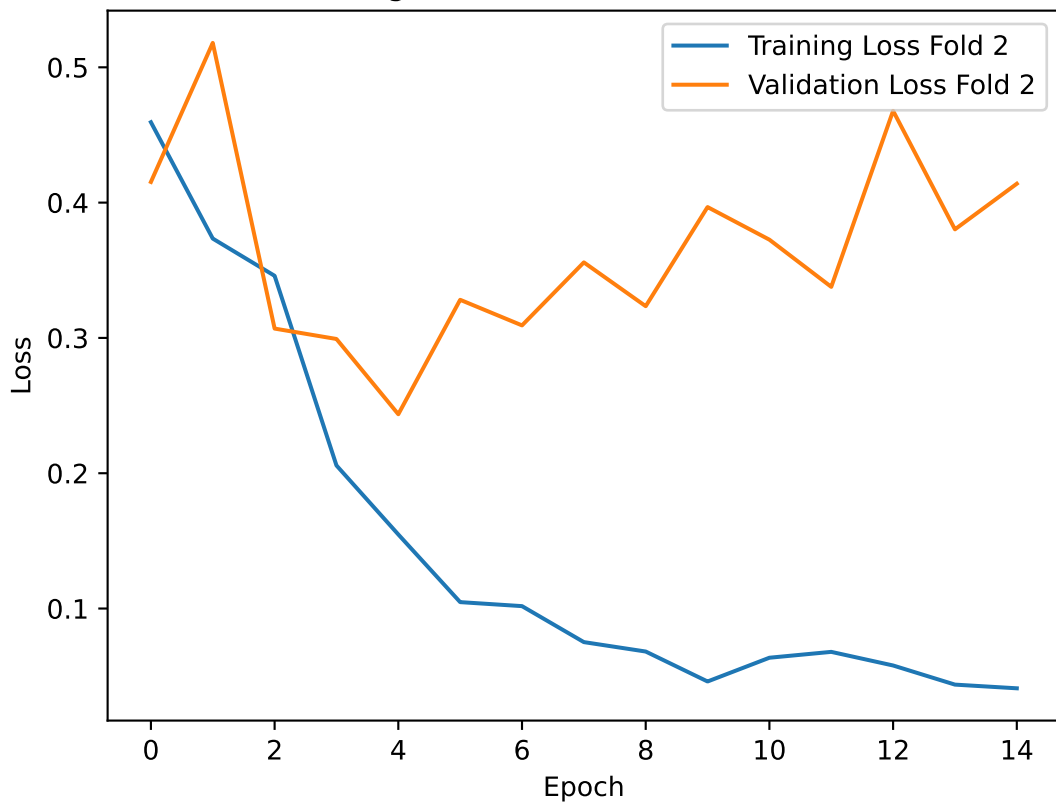
Best Trial Score for Fold 1: 0.8083332777023315



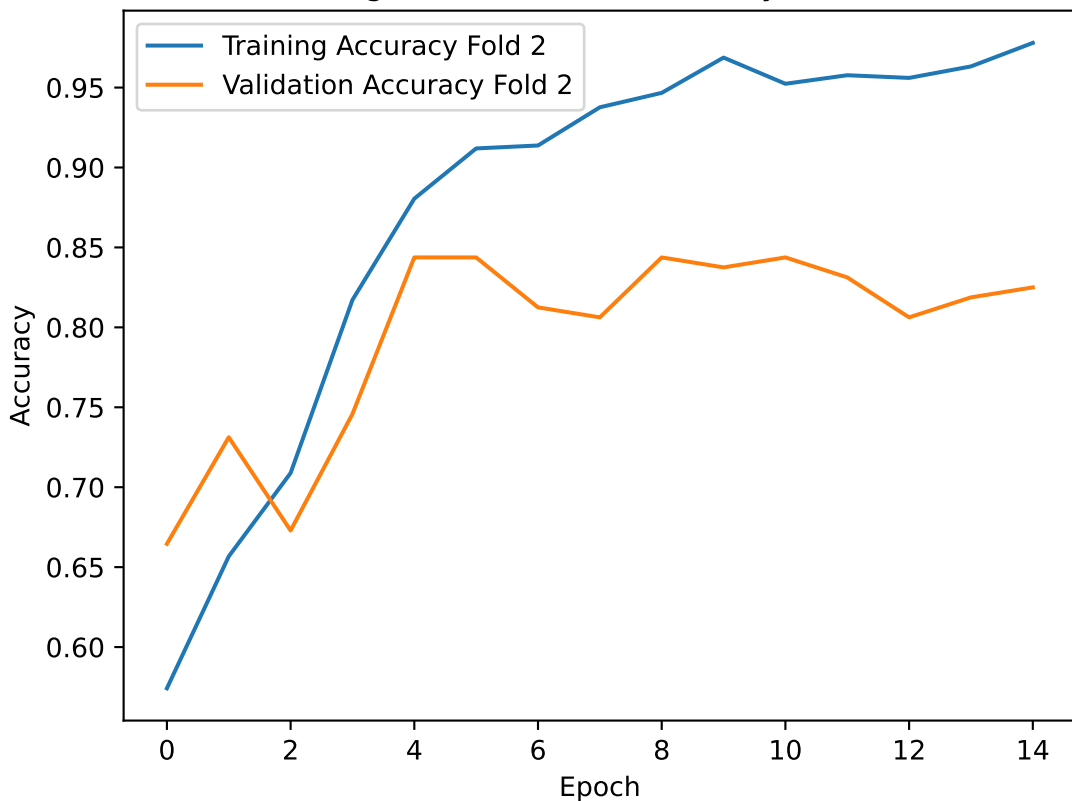
Validation F1-Score for fold 2: 0.05947955390334572

0.05947955390334572

Training and Validation Loss for Fold 2



Training and Validation Accuracy for Fold 2



Best Hyperparameters for Fold 2:

`{'conv_filters': 160, 'conv_activation': 'relu', 'unitsD1': 416, 'unitsD2': 160}`

Best Trial Score for Fold 2: 0.8500000238418579

Search: Running Trial #1

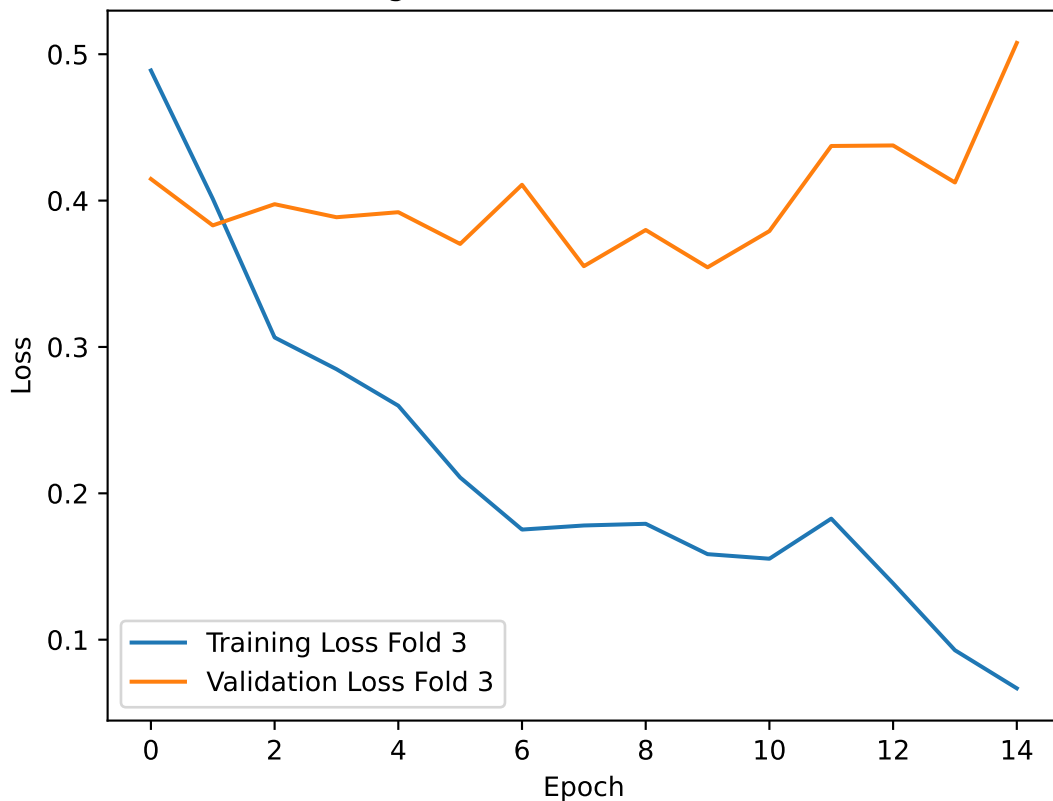
| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 128   | 128               | conv_filters    |
| relu  | relu              | conv_activation |
| 32    | 32                | unitsD1         |
| 32    | 32                | unitsD2         |

Epoch 1/15  
[1m 1/17][0m [32m][0m[37m-----[0m [1m10:01][0m 38s/step - accuracy: 0.4688 - loss: 0.5091  
[32m][0m[37m-----[0m [1m3s][0m 253ms/step - accuracy: 0.4688 - loss: 0.5105  
[32m][0m[37m-----[0m [1m3s][0m 251ms/step - accuracy: 0.4826 - loss: 0.5077  
[32m][0m[37m-----[0m [1m3s][0m 251ms/step - accuracy: 0.5065 - loss: 0.4920  
[32m][0m[37m-----[0m [1m3s][0m 252ms/step - accuracy: 0.5215 - loss: 0.4828  
[32m][0m[37m-----[0m [1m2s][0m 252ms/step - accuracy: 0.5257 - loss: 0.4817  
[32m][0m[37m-----[0m [1m2s][0m 253ms/step - accuracy: 0.5227 - loss: 0.4838  
[32m][0m[37m-----[0m [1m2s][0m 252ms/step - accuracy: 0.5276 - loss: 0.4839  
[32m][0m[37m-----[0m [1m2s][0m 252ms/step - accuracy: 0.5319 - loss: 0.4840  
[32m][0m[37m-----[0m [1m1s][0m 252ms/step - accuracy: 0.5353 - loss: 0.4844  
[32m][0m[37m-----[0m [1m1s][0m 252ms/step - accuracy: 0.5385 - loss: 0.4843  
[32m][0m[37m-----[0m [1m1s][0m 252ms/step - accuracy: 0.5399 - loss: 0.4847  
[32m][0m[37m-----[0m [1m1s][0m 252ms/step - accuracy: 0.5399 - loss: 0.4852  
[32m][0m[37m-----[0m [1m0s][0m 252ms/step - accuracy: 0.5404 - loss: 0.4855  
[32m][0m[37m-----[0m [1m0s][0m 252ms/step - accuracy: 0.5415 - loss: 0.4853  
[32m][0m[37m-----[0m [1m0s][0m 252ms/step - accuracy: 0.5426 - loss: 0.4850  
[32m][0m[37m[0m [1m0s][0m 832ms/step - accuracy: 0.5441 - loss: 0.4846  
[32m][0m[37m[0m [1m71s][0m 2s/step - accuracy: 0.5454 - loss: 0.4843 - val\_accuracy: 0.6708 - val\_loss: 0.4723  
Epoch 2/15  
[1m 1/17][0m [32m][0m[37m-----[0m [1m4s][0m 300ms/step - accuracy: 0.5625 - loss: 0.5185  
[32m][0m[37m-----[0m [1m3s][0m 247ms/step - accuracy: 0.5234 - loss: 0.5638  
[32m][0m[37m-----[0m [1m3s][0m 245ms/step - accuracy: 0.5295 - loss: 0.5601  
[32m][0m[37m-----[0m [1m3s][0m 245ms/step - accuracy: 0.5475 - loss: 0.5393  
[32m][0m[37m-----[0m [1m2s][0m 244ms/step - accuracy: 0.5580 - loss: 0.5244  
[32m][0m[37m-----[0m [1m2s][0m 245ms/step - accuracy: 0.5622 - loss: 0.5184  
[32m][0m[37m-----[0m [1m2s][0m 245ms/step - accuracy: 0.5687 - loss: 0.5097  
[32m][0m[37m-----[0m [1m2s][0m 245ms/step - accuracy: 0.5781 - loss: 0.4976  
[32m][0m[37m-----[0m [1m1s][0m 246ms/step - accuracy: 0.5841 - loss: 0.4892  
[32m][0m[37m-----[0m [1m1s][0m 246ms/step - accuracy: 0.5888 - loss: 0.4828  
[32m][0m[37m-----[0m [1m1s][0m 246ms/step - accuracy: 0.5926 - loss: 0.4776  
[32m][0m[37m-----[0m [1m1s][0m 246ms/step - accuracy: 0.5942 - loss: 0.4737  
[32m][0m[37m-----[0m [1m0s][0m 246ms/step - accuracy: 0.5953 - loss: 0.4706  
[32m][0m[37m-----[0m [1m0s][0m 246ms/step - accuracy: 0.5954 - loss: 0.4683  
[32m][0m[37m-----[0m [1m0s][0m 246ms/step - accuracy: 0.5960 - loss: 0.4660  
[32m][0m[37m-----[0m [1m0s][0m 246ms/step - accuracy: 0.5969 - loss: 0.4636  
[32m][0m[37m[0m [1m0s][0m 237ms/step - accuracy: 0.5985 - loss: 0.4613  
[32m][0m[37m[0m [1m6s][0m 337ms/step - accuracy: 0.5999 - loss: 0.4593 - val\_accuracy: 0.7083 - val\_loss: 0.4078  
Epoch 3/15

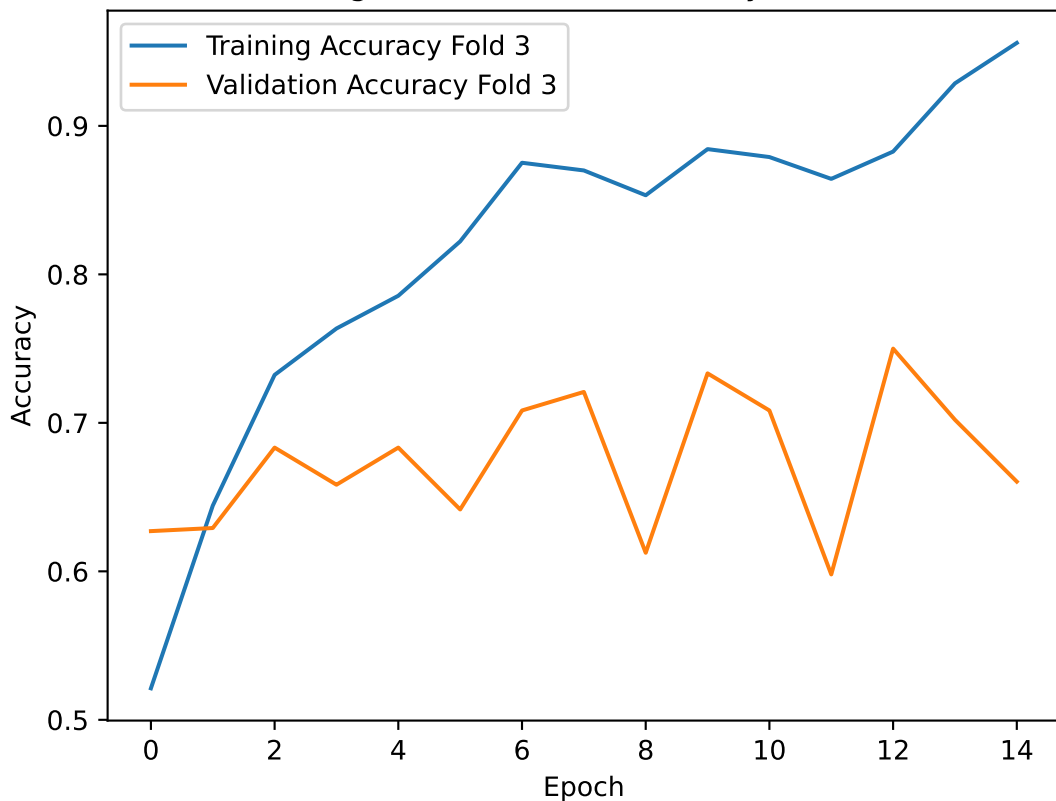
Validation F1-Score for fold 3: 0.10218978102189781

0.10218978102189781

Training and Validation Loss for Fold 3



Training and Validation Accuracy for Fold 3



Best Hyperparameters for Fold 3:

```
{'conv_filters': 160, 'conv_activation': 'relu', 'unitsD1': 32, 'unitsD2': 160}
```

Best Trial Score for Fold 3: 0.7875000238418579

Search: Running Trial #1

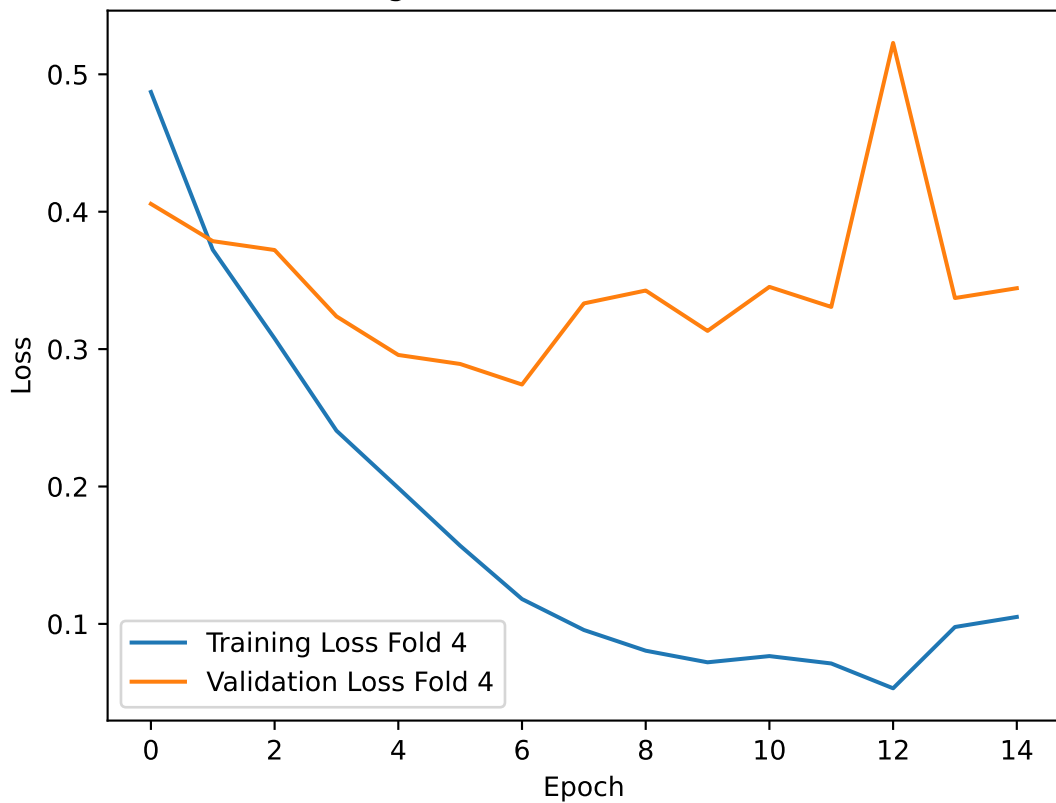
| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 224   | 224               | conv_filters    |
| relu  | relu              | conv_activation |
| 288   | 288               | unitsD1         |
| 32    | 32                | unitsD2         |

Epoch 1/15  
[1m 1/17][0m [32m][0m[37m[0m [1m9:49][0m 37s/step - accuracy: 0.3438 - loss: 0.5434[1m 2/17][0m [32m][0m[37m[0m [1m3s][0m 266ms/step - accuracy: 0.3672 - loss: 0.5384[1m 3/17][0m [32m][0m[37m[0m [1m3s][0m 264ms/step - accuracy: 0.3941 - loss: 0.5293[1m 4/17][0m [32m][0m[37m[0m [1m3s][0m 263ms/step - accuracy: 0.4225 - loss: 0.5147[1m 5/17][0m [32m][0m[37m[0m [1m3s][0m 263ms/step - accuracy: 0.4480 - loss: 0.5001[1m 6/17][0m [32m][0m[37m[0m [1m2s][0m 263ms/step - accuracy: 0.4610 - loss: 0.4999[1m 7/17][0m [32m][0m[37m[0m [1m2s][0m 263ms/step - accuracy: 0.4685 - loss: 0.5025[1m 8/17][0m [32m][0m[37m[0m [1m2s][0m 263ms/step - accuracy: 0.4812 - loss: 0.4985[1m 9/17][0m [32m][0m[37m[0m [1m2s][0m 263ms/step - accuracy: 0.4918 - loss: 0.4953[1m10/17][0m [32m][0m[37m[0m [1m1s][0m 263ms/step - accuracy: 0.5004 - loss: 0.4933[1m11/17][0m [32m][0m[37m[0m [1m1s][0m 262ms/step - accuracy: 0.5079 - loss: 0.4910[1m12/17][0m [32m][0m[37m[0m [1m1s][0m 262ms/step - accuracy: 0.5140 - loss: 0.4895[1m13/17][0m [32m][0m[37m[0m [1m1s][0m 262ms/step - accuracy: 0.5179 - loss: 0.4890[1m14/17][0m [32m][0m[37m[0m [1m0s][0m 262ms/step - accuracy: 0.5209 - loss: 0.4888[1m15/17][0m [32m][0m[37m[0m [1m0s][0m 262ms/step - accuracy: 0.5246 - loss: 0.4883[1m16/17][0m [32m][0m[37m[0m [1m0s][0m 262ms/step - accuracy: 0.5280 - loss: 0.4879[1m17/17][0m [32m][0m[37m[0m [1m0s][0m 764ms/step - accuracy: 0.5305 - loss: 0.4877[1m17/17][0m [32m][0m[37m[0m [1m68s][0m 2s/step - accuracy: 0.5328 - loss: 0.4875 - val\_accuracy: 0.5417 - val\_loss: 0.4627  
Epoch 2/15  
[1m 1/17][0m [32m][0m[37m[0m [1m5s][0m 319ms/step - accuracy: 0.4688 - loss: 0.5430[1m 2/17][0m [32m][0m[37m[0m [1m3s][0m 258ms/step - accuracy: 0.4609 - loss: 0.5453[1m 3/17][0m [32m][0m[37m[0m [1m3s][0m 258ms/step - accuracy: 0.4878 - loss: 0.5281[1m 4/17][0m [32m][0m[37m[0m [1m3s][0m 258ms/step - accuracy: 0.5163 - loss: 0.5041[1m 5/17][0m [32m][0m[37m[0m [1m3s][0m 257ms/step - accuracy: 0.5393 - loss: 0.4871[1m 6/17][0m [32m][0m[37m[0m [1m2s][0m 258ms/step - accuracy: 0.5492 - loss: 0.4790[1m 7/17][0m [32m][0m[37m[0m [1m2s][0m 258ms/step - accuracy: 0.5562 - loss: 0.4711[1m 8/17][0m [32m][0m[37m[0m [1m2s][0m 258ms/step - accuracy: 0.5677 - loss: 0.4585[1m 9/17][0m [32m][0m[37m[0m [1m2s][0m 258ms/step - accuracy: 0.5780 - loss: 0.4469[1m10/17][0m [32m][0m[37m[0m [1m1s][0m 258ms/step - accuracy: 0.5867 - loss: 0.4391[1m11/17][0m [32m][0m[37m[0m [1m1s][0m 258ms/step - accuracy: 0.5946 - loss: 0.4316[1m12/17][0m [32m][0m[37m[0m [1m1s][0m 258ms/step - accuracy: 0.6017 - loss: 0.4253[1m13/17][0m [32m][0m[37m[0m [1m1s][0m 258ms/step - accuracy: 0.6068 - loss: 0.4208[1m14/17][0m [32m][0m[37m[0m [1m0s][0m 258ms/step - accuracy: 0.6110 - loss: 0.4175[1m15/17][0m [32m][0m[37m[0m [1m0s][0m 258ms/step - accuracy: 0.6154 - loss: 0.4138[1m16/17][0m [32m][0m[37m[0m [1m0s][0m 258ms/step - accuracy: 0.6192 - loss: 0.4106[1m17/17][0m [32m][0m[37m[0m [1m0s][0m 248ms/step - accuracy: 0.6227 - loss: 0.4077[1m17/17][0m [32m][0m[37m[0m [1m7s][0m 391ms/step - accuracy: 0.6258 - loss: 0.4052 - val\_accuracy: 0.5604 - val\_loss: 0.4227  
Epoch 3/15

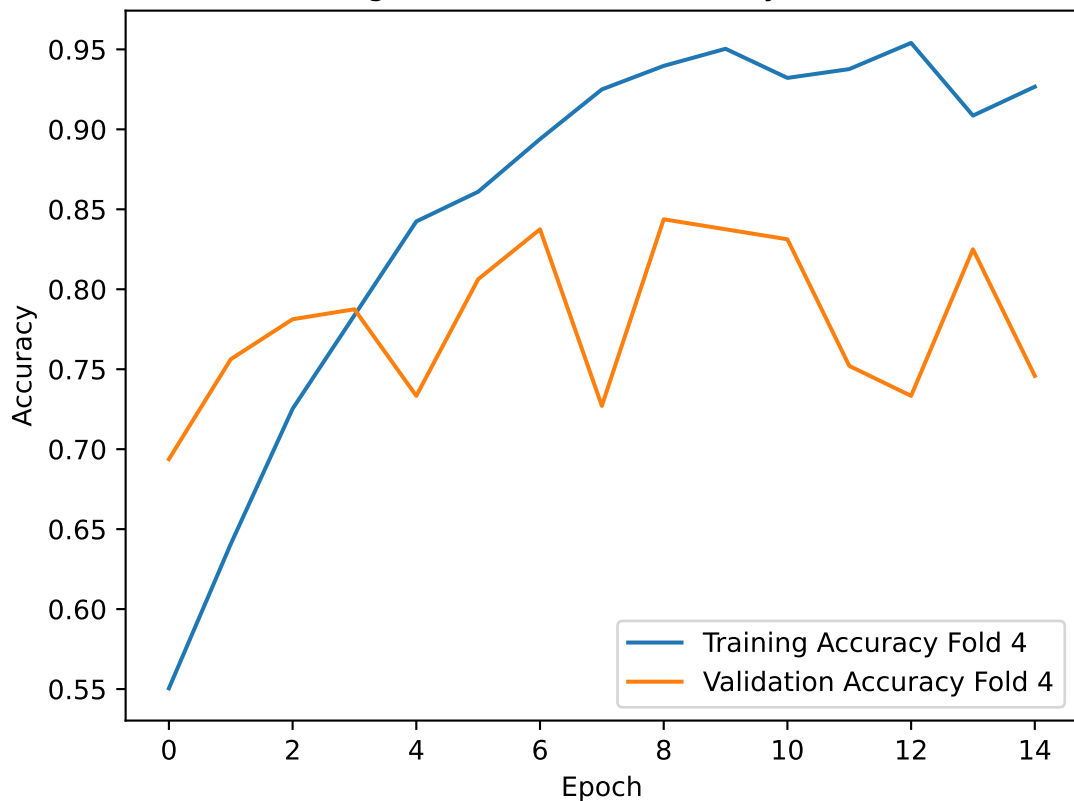
Validation F1-Score for fold 4: 0.1494661921708185

0.1494661921708185

Training and Validation Loss for Fold 4



Training and Validation Accuracy for Fold 4



Best Hyperparameters for Fold 4:

`{'conv_filters': 160, 'conv_activation': 'relu', 'unitsD1': 160, 'unitsD2': 160}`

Best Trial Score for Fold 4: 0.875

Search: Running Trial #1

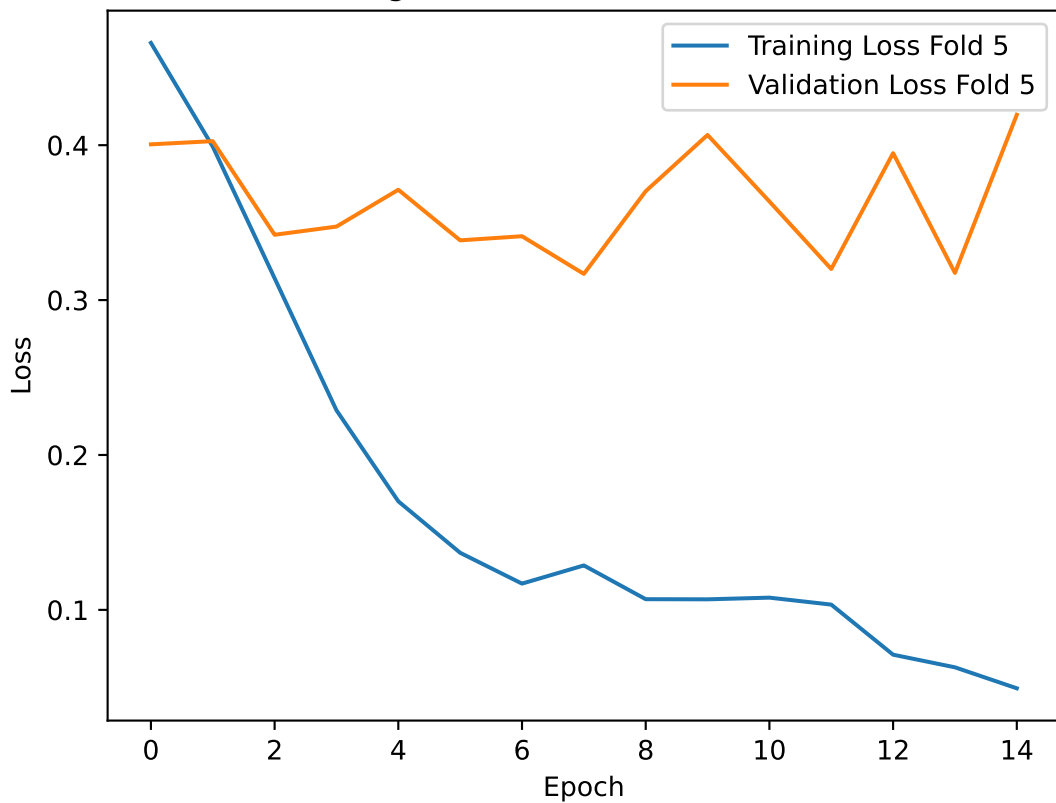
| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 128   | 128               | conv_filters    |
| relu  | relu              | conv_activation |
| 288   | 288               | unitsD1         |
| 160   | 160               | unitsD2         |

Epoch 1/15  
[1m 1/17][0m [32m][0m][37m][0m [1m9:49][0m 37s/step - accuracy: 0.6250 - loss: 0.5132][1m 2/17][0m [32m][0m][37m][0m [1m3s][0m 261ms/step - accuracy: 0.5547 - loss: 0.5346][1m 3/17][0m [32m][0m][37m][0m [1m3s][0m 260ms/step - accuracy: 0.5156 - loss: 0.5433][1m 4/17][0m [32m][0m][37m][0m [1m3s][0m 260ms/step - accuracy: 0.5059 - loss: 0.5400][1m 5/17][0m [32m][0m][37m][0m [1m3s][0m 260ms/step - accuracy: 0.5072 - loss: 0.5361][1m 6/17][0m [32m][0m][37m][0m [1m2s][0m 260ms/step - accuracy: 0.5060 - loss: 0.5387][1m 7/17][0m [32m][0m][37m][0m [1m2s][0m 260ms/step - accuracy: 0.5039 - loss: 0.5408][1m 8/17][0m [32m][0m][37m][0m [1m2s][0m 261ms/step - accuracy: 0.5092 - loss: 0.5358][1m 9/17][0m [32m][0m][37m][0m [1m2s][0m 261ms/step - accuracy: 0.5163 - loss: 0.5293][1m10/17][0m [32m][0m][37m][0m [1m1s][0m 261ms/step - accuracy: 0.5219 - loss: 0.5245][1m11/17][0m [32m][0m][37m][0m [1m1s][0m 261ms/step - accuracy: 0.5269 - loss: 0.5202][1m12/17][0m [32m][0m][37m][0m [1m1s][0m 261ms/step - accuracy: 0.5300 - loss: 0.5172][1m13/17][0m [32m][0m][37m][0m [1m1s][0m 261ms/step - accuracy: 0.5324 - loss: 0.5153][1m14/17][0m [32m][0m][37m][0m [1m0s][0m 261ms/step - accuracy: 0.5339 - loss: 0.5139][1m15/17][0m [32m][0m][37m][0m [1m0s][0m 261ms/step - accuracy: 0.5359 - loss: 0.5122][1m16/17][0m [32m][0m][37m][0m [1m0s][0m 261ms/step - accuracy: 0.5373 - loss: 0.5111][1m17/17][0m [32m][0m][37m][0m [1m0s][0m 768ms/step - accuracy: 0.5382 - loss: 0.5105][0m [32m][0m][37m][0m [1m68s][0m 2s/step - accuracy: 0.5390 - loss: 0.5100 - val\_accuracy: 0.7500 - val\_loss: 0.3663  
Epoch 2/15  
[1m 1/17][0m [32m][0m][37m][0m [1m5s][0m 329ms/step - accuracy: 0.6250 - loss: 0.4928][1m 2/17][0m [32m][0m][37m][0m [1m3s][0m 257ms/step - accuracy: 0.5625 - loss: 0.5359][1m 3/17][0m [32m][0m][37m][0m [1m3s][0m 256ms/step - accuracy: 0.5625 - loss: 0.5281][1m 4/17][0m [32m][0m][37m][0m [1m3s][0m 257ms/step - accuracy: 0.5762 - loss: 0.5083][1m 5/17][0m [32m][0m][37m][0m [1m3s][0m 258ms/step - accuracy: 0.5872 - loss: 0.4925][1m 6/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.5900 - loss: 0.4872][1m 7/17][0m [32m][0m][37m][0m [1m2s][0m 258ms/step - accuracy: 0.5918 - loss: 0.4823][1m 8/17][0m [32m][0m][37m][0m [1m2s][0m 258ms/step - accuracy: 0.5994 - loss: 0.4717][1m 9/17][0m [32m][0m][37m][0m [1m2s][0m 258ms/step - accuracy: 0.6080 - loss: 0.4614][1m10/17][0m [32m][0m][37m][0m [1m1s][0m 259ms/step - accuracy: 0.6150 - loss: 0.4535][1m11/17][0m [32m][0m][37m][0m [1m1s][0m 258ms/step - accuracy: 0.6216 - loss: 0.4456][1m12/17][0m [32m][0m][37m][0m [1m1s][0m 259ms/step - accuracy: 0.6258 - loss: 0.4395][1m13/17][0m [32m][0m][37m][0m [1m1s][0m 259ms/step - accuracy: 0.6280 - loss: 0.4355][1m14/17][0m [32m][0m][37m][0m [1m0s][0m 259ms/step - accuracy: 0.6300 - loss: 0.4322][1m15/17][0m [32m][0m][37m][0m [1m0s][0m 259ms/step - accuracy: 0.6319 - loss: 0.4290][1m16/17][0m [32m][0m][37m][0m [1m0s][0m 259ms/step - accuracy: 0.6327 - loss: 0.4270][1m17/17][0m [32m][0m][37m][0m [1m0s][0m 250ms/step - accuracy: 0.6330 - loss: 0.4252][0m [32m][0m][37m][0m [1m5s][0m 304ms/step - accuracy: 0.6333 - loss: 0.4237 - val\_accuracy: 0.6833 - val\_loss: 0.3631  
Epoch 3/15

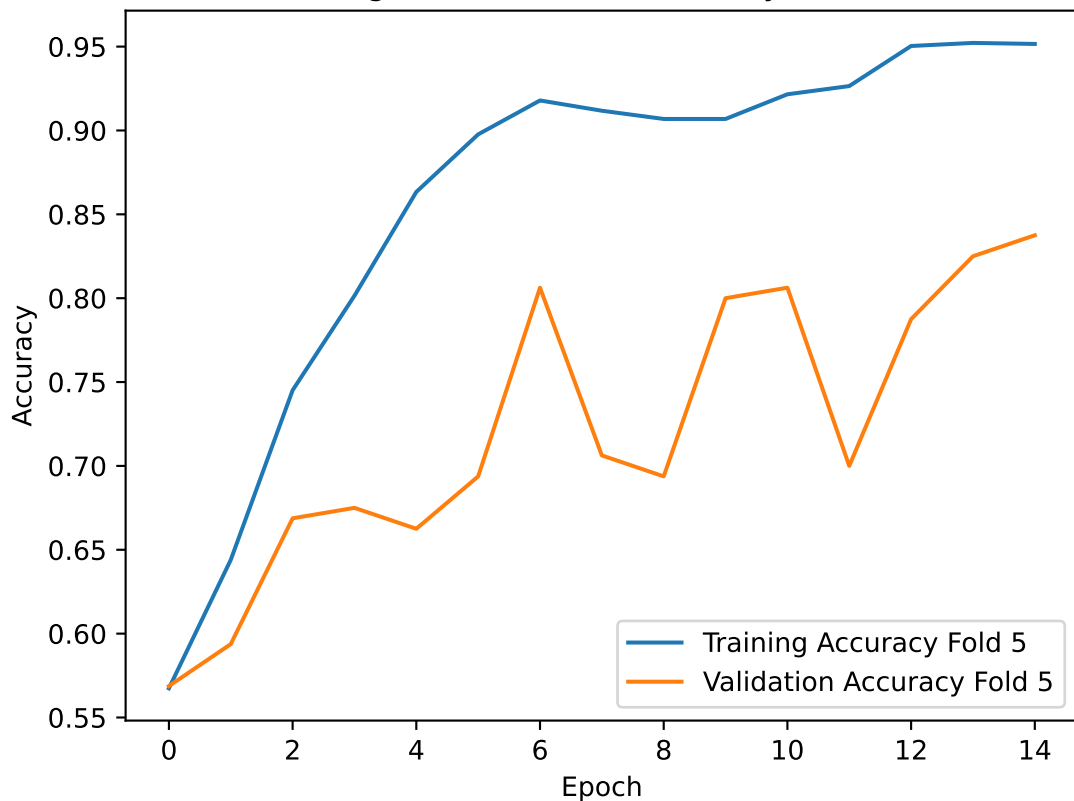
Validation F1-Score for fold 5: 0.08239700374531836

0.08239700374531836

Training and Validation Loss for Fold 5



Training and Validation Accuracy for Fold 5



Best Hyperparameters for Fold 5:

```
{'conv_filters': 224, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 32}
```

Best Trial Score for Fold 5: 0.8187500238418579

Search: Running Trial #1

| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 224   | 224               | conv_filters    |
| relu  | relu              | conv_activation |
| 288   | 288               | unitsD1         |
| 32    | 32                | unitsD2         |

Epoch 1/15  
[1m 1/17][0m [32m][0m][37m-----][0m [1m9:52][0m 37s/step - accuracy: 0.5938 - loss: 0.5031  
[32m][0m][37m-----][0m [1m3s][0m 260ms/step - accuracy: 0.5234 - loss: 0.5098  
[32m][0m][37m-----][0m [1m3s][0m 261ms/step - accuracy: 0.5191 - loss: 0.5081  
[32m][0m][37m-----][0m [1m3s][0m 262ms/step - accuracy: 0.5299 - loss: 0.4954  
[32m][0m][37m-----][0m [1m3s][0m 262ms/step - accuracy: 0.5390 - loss: 0.4873  
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.5368 - loss: 0.4904  
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.5366 - loss: 0.4928  
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.5433 - loss: 0.4877  
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.5458 - loss: 0.4855  
[32m][0m][37m-----][0m [1m1s][0m 262ms/step - accuracy: 0.5484 - loss: 0.4836  
[32m][0m][37m-----][0m [1m1s][0m 262ms/step - accuracy: 0.5515 - loss: 0.4817  
[32m][0m][37m-----][0m [1m1s][0m 262ms/step - accuracy: 0.5542 - loss: 0.4803  
[32m][0m][37m-----][0m [1m1s][0m 262ms/step - accuracy: 0.5557 - loss: 0.4791  
[32m][0m][37m-----][0m [1m0s][0m 262ms/step - accuracy: 0.5575 - loss: 0.4783  
[32m][0m][37m-----][0m [1m0s][0m 262ms/step - accuracy: 0.5591 - loss: 0.4775  
[32m][0m][37m-----][0m [1m0s][0m 262ms/step - accuracy: 0.5599 - loss: 0.4773  
[32m][0m][37m][0m [1m0s][0m 1s/step - accuracy: 0.5612 - loss: 0.4769  
[32m][0m][37m][0m [1m74s][0m 2s/step - accuracy: 0.5624 - loss: 0.4765 - val\_accuracy: 0.6687 - val\_loss: 0.4282  
Epoch 2/15  
[1m 1/17][0m [32m][0m][37m-----][0m [1m5s][0m 352ms/step - accuracy: 0.6250 - loss: 0.4229  
[32m][0m][37m-----][0m [1m3s][0m 252ms/step - accuracy: 0.5859 - loss: 0.4419  
[32m][0m][37m-----][0m [1m3s][0m 255ms/step - accuracy: 0.5677 - loss: 0.4587  
[32m][0m][37m-----][0m [1m3s][0m 255ms/step - accuracy: 0.5664 - loss: 0.4579  
[32m][0m][37m-----][0m [1m3s][0m 255ms/step - accuracy: 0.5631 - loss: 0.4591  
[32m][0m][37m-----][0m [1m2s][0m 256ms/step - accuracy: 0.5569 - loss: 0.4652  
[32m][0m][37m-----][0m [1m2s][0m 256ms/step - accuracy: 0.5597 - loss: 0.4629  
[32m][0m][37m-----][0m [1m2s][0m 255ms/step - accuracy: 0.5678 - loss: 0.4548  
[32m][0m][37m-----][0m [1m2s][0m 255ms/step - accuracy: 0.5773 - loss: 0.4462  
[32m][0m][37m-----][0m [1m1s][0m 256ms/step - accuracy: 0.5855 - loss: 0.4398  
[32m][0m][37m-----][0m [1m1s][0m 256ms/step - accuracy: 0.5929 - loss: 0.4334  
[32m][0m][37m-----][0m [1m1s][0m 256ms/step - accuracy: 0.5993 - loss: 0.4281  
[32m][0m][37m-----][0m [1m1s][0m 256ms/step - accuracy: 0.6046 - loss: 0.4241  
[32m][0m][37m-----][0m [1m0s][0m 256ms/step - accuracy: 0.6083 - loss: 0.4215  
[32m][0m][37m-----][0m [1m0s][0m 256ms/step - accuracy: 0.6114 - loss: 0.4193  
[32m][0m][37m-----][0m [1m0s][0m 256ms/step - accuracy: 0.6135 - loss: 0.4178  
[32m][0m][37m][0m [1m0s][0m 247ms/step - accuracy: 0.6158 - loss: 0.4164  
[32m][0m][37m][0m [1m5s][0m 300ms/step - accuracy: 0.6179 - loss: 0.4152 - val\_accuracy: 0.6625 - val\_loss: 0.3675  
Epoch 3/15

Average Validation F1-Score across all folds:

0.08628226374403365

Average Validation MSE across all folds:

0.36019203583399456

Average Validation Accuracy across all folds:

0.740972228050232

### Best Hyperparameters Found:

Best Fold: 4

Best Trial Score: 0.875

Best Hyperparameters: {'conv\_filters': 160, 'conv\_activation': 'relu', 'units': 100, 'activation': 'tanh', 'learning\_rate': 0.001, 'epochs': 100, 'batch\_size': 32, 'optimizer': 'adam', 'loss': 'categorical\_crossentropy', 'metrics': ['accuracy']}